SUHAS NAGARAJ

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EDUCATION

Master of Engineering in Robotics & Artificial Intelligence [CGPA: 4.0 / 4.0] — **August 2023 - May 2025**
University Of Maryland, College Park [Phi Kappa Phi Honor Society Member] — College Park, United States

Bachelor of Engineering in Mechanical Engineering [CGPA: 9.3 / 10.0] — **August 2017 - July 2021**
Ramaiah Institute of Technology — Bengaluru, India

PROFESSIONAL EXPERIENCE

Teaching Assistant [MSML605: Computing Systems for Machine Learning] — **January 2025 – May 2025**
College of Computer, Mathematical & Natural Sciences, University of Maryland — College Park, USA

- **Guiding 60 graduate students** in Python fundamentals and machine learning hardware concepts.
- **Instructing** students on effectively leveraging the University's central HPC cluster for model training, reducing average computational time by **90%** and enabling more advanced experiments.
- **Collaborating** with Dr. Jerry Wu to revamp course content resulting in a **20% increase** in class participation and engagement.

Research Intern — **May 2024 – Present**
Intelligent Inclusive Interaction Design (I3D) Lab, Indian Institute of Science (IISc) — Bengaluru, India

- **Developed** a novel 3D reconstruction algorithm integrating LiDAR and Gaussian Splatting SLAM, improving localization accuracy by **200%**.
- **Reduced** computational overhead by **35%**; validated the system in a simulation environment and on a real TurtleBot.
- **Published** results at ICRA (among the **39%** of papers accepted).

Engineering Intern — **October 2022 – March 2023**
DiFACTO Robotics and Automation Pvt. Ltd — Bengaluru, India

- **Designed** automation parts using SolidWorks, **performed** industrial process simulations with FANUC RoboGuide across **5 projects**.
- **Programmed ABB** (RAPID) and **FANUC** (KAREL) robots in both offline and online environments,
- **Developed** PLC programming and DeviceNet interfacing skills to control robot cells using Mitsubishi PLC.

Technical Consultant — **August 2021 – March 2022**
U-Solar Clean Energy Solutions Pvt. Ltd. — Bengaluru, India

- **Managed** a portfolio of **50+ client accounts** and **10 partner relationships**, contributing to a **20% increase** in customer retention.
- **Spearheaded** market and policy research initiatives, resulting in a **15% growth** in market reach and an expanded project pipeline.
- **Collaborated** with cross-functional teams to devise technical strategies, driving **25% higher lead generation** and **overall project success**.

PUBLICATION

3D Reconstruction via Camera-Lidar (2D) Fusion for Mobile Robots: A Gaussian Splatting Approach
*Ajay Kumar Sandula, *Shriram Damodaran, ****Suhas Nagaraj***, *Debasish Ghose and Pradipta Biswas [* are equally contributing authors]*
IEEE International Conference on Robotics and Automation (ICRA) 2025

SKILLS

- **Technical Expertise:** ROS1/ROS2, Computer Vision, Machine Learning, Neural Networks, Natural Language Processing, Simulation, Robot Programming, SLAM, 3D reconstruction, Data Analytics, Sensor Fusion, Multi-threading, CUDA, Robot Hardware and Modelling, Could Computing, MLOps, Containerization, Version Control, Optimization, Control theory, Product Design, Rapid Prototyping, PLC Programming.
- **Programming Languages:** Python, C++, MATLAB, RAPID (ABB), KAREL (FANUC).
- **Libraries, Frameworks and Tools:** PyTorch, Cuda, Tensorflow, Keras, ONNX, Huggingface, OpenCV, Open3D, PCL, Yolo, Cartographer, NumPy, pandas, scikit-learn, NLTK, Kubernetes, Docker, AWS, Azure, GCP, Linux, GitHub.
- **Others:** MATLAB Simulink, FANUC ROBOGUIDE, ABB Robot Studio, CimStation Robotics, SolidWorks, AutoCAD, Fusion 360, CATIA.

PROJECTS [www.suhasnagaraj.com/projects]

- **Adaptive Text-to-Command Translation for Robotic Navigation:** Fine-tuned T5-Small [w/wo LoRA] model to convert natural language commands into structured navigation plans for TurtleBot3 in ROS2 and Gazebo, achieving 100% test accuracy during deployment.
- **ASD Prediction Using Eye Tracking Data:** Developed a novel deep learning approach (CNN, LSTM, Transformers) for classifying Autism Spectrum Disorder using eye-tracking data, achieving 88.9% accuracy.
- **Automated Guided Vehicle:** Designed autonomous navigation with lane following, stop sign detection, and dynamic obstacle avoidance for mobile robots using ROS2 and OpenCV in simulation and real-world setups, using Turtlebot3 Waffle.
- **Maze Navigation with Object Detection:** Developed ROS2-based system for maze traversal using Aruco markers and object detection
- **SLAM-Based Waypoint Navigation:** Designed an autonomous waypoint navigation system for TurtleBot3 using SLAM, TF tree localization.
- **Deep Reinforcement Learning:** Applied Deep Q-Networks (DQN) for obstacle avoidance and autonomous navigation of mobile robots.
- **Inverse Kinematics & Machine Tending:** Focused on IK-based control of ABB IRB 1600 in Gazebo with ROS2, enabling waypoint-based robot motion.
- **Path Planning Algorithms:** Developed Dijkstra's, A*, RRT, RRT*, Bidirectional A*, Potential Field and Real Time RRT* algorithms for mobile robot navigation, implemented the solutions on a Turtlebot3 Waffle robot in both simulation and real-world setups.
- **Advanced Control strategies:** Implemented feedback linearization & sliding mode (quadcopter), LQR/LQG (dual-pendulum cart).